

FIR Filter Architectures: Pipelining and Resource Analysis

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1 Objective

To implement FIR filters using multiple architectures and compare:

- Direct, Optimized, and Genvar-based designs
- Their pipelined versions
- FIR IP core

2 Theory

FIR filter:

$$y[n] = \sum_{k=0}^{N-1} h[k]x[n-k]$$

3 Design Implementations

3.1 Direct Form FIR

The Direct Form FIR filter is implemented using a sequence of delay elements, multipliers, and an adder tree. Each input sample is delayed and multiplied with corresponding filter coefficients, and the results are accumulated to produce the output.

This architecture is straightforward but suffers from a long combinational path due to the cascaded addition of all multiplier outputs.

Pipelined Direct Form:

In the pipelined version, registers are inserted between stages of the adder chain. This reduces the critical path delay by breaking the long combinational path into smaller segments.

- Registers are placed after partial sums
- Each stage computes a portion of the accumulation
- Latency increases due to added pipeline stages
- Throughput improves significantly

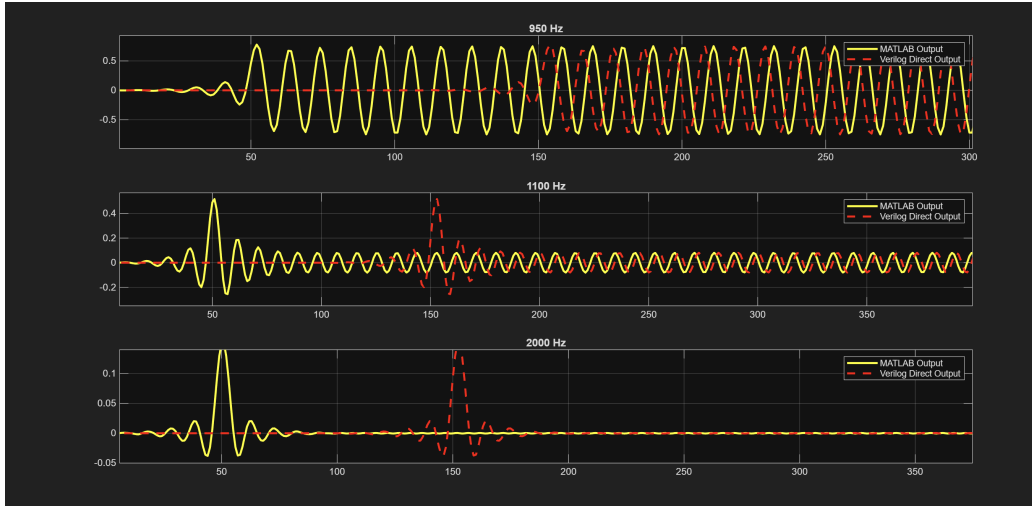


Figure 1: Direct FIR Output (Pipelined)

Resource Comparison (Direct Form):

Flow Summary	
<<Filter>>	
Flow Status	Successful - Wed Mar 25 23:19:00 2026
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	exp5
Top-level Entity Name	fir_direct
Family	Cyclone V
Device	5CGTFD5C5F2717
Timing Models	Final
Logic utilization (in ALMs)	N/A
Total registers	1616
Total pins	49
Total virtual pins	0
Total block memory bits	0
Total DSP Blocks	99
Total HSSI RX PCSs	0
Total HSSI PMA RX Deserializers	0
Total HSSI TX PCSs	0
Total HSSI PMA TX Serializers	0
Total PLLs	0
Total DLLs	0

Figure 2: Resource Utilization Comparison: Direct

Flow Summary	
<<Filter>>	
Flow Status	Successful - Wed Mar 25 23:15:14 2026
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	exp5
Top-level Entity Name	fir_direct
Family	Cyclone V
Device	5CGTFD5C5F2717
Timing Models	Final
Logic utilization (in ALMs)	N/A
Total registers	6384
Total pins	50
Total virtual pins	0
Total block memory bits	0
Total DSP Blocks	100
Total HSSI RX PCSs	0
Total HSSI PMA RX Deserializers	0
Total HSSI TX PCSs	0
Total HSSI PMA TX Serializers	0
Total PLLs	0
Total DLLs	0

Figure 3: Resource Utilization Comparison: Pipelined

Fmax Comparison

Slow 1100mV 100C Model Fmax Summary				
<<Filter>>				
	Fmax	Restricted Fmax	Clock Name	Note
1	62.38 MHz	62.38 MHz	clk	

Figure 4: Fmax: Direct

Slow 1100mV 100C Model Fmax Summary				
<<Filter>>				
	Fmax	Restricted Fmax	Clock Name	Note
1	125.53 MHz	125.53 MHz	clk	

Figure 5: Fmax: Pipelined

Multi Corner Timing Analysis Comparison

Multicorner Timing Analysis Summary						
<<Filter>>						
	Clock	Setup	Hold	Recovery	Removal	Minimum Pulse Width
1	▼ Worst-case Slack	4.949	0.016	N/A	N/A	6.687
1	clk	4.949	0.016	N/A	N/A	6.687
2	▼ Design-wide TNS	0.0	0.0	0.0	0.0	0.0
1	clk	0.000	0.000	N/A	N/A	0.000

Figure 6: Multi Corner Timing Analysis: Direct

Multicorner Timing Analysis Summary						
<<Filter>>						
	Clock	Setup	Hold	Recovery	Removal	Minimum Pulse Width
1	▼ Worst-case Slack	6.607	0.093	N/A	N/A	6.679
1	clk	6.607	0.093	N/A	N/A	6.679
2	▼ Design-wide TNS	0.0	0.0	0.0	0.0	0.0
1	clk	0.000	0.000	N/A	N/A	0.000

Figure 7: Multi Corner Timing Analysis: Pipelined

3.2 Optimized FIR

The optimized FIR implementation improves efficiency by restructuring computations, typically reducing redundant additions and improving data flow.

This design minimizes hardware usage while maintaining correct functionality.

Pipelined Optimized FIR:

Pipeline registers are introduced at carefully chosen locations to balance delay across stages.

- Balanced pipeline stages reduce delay variation
- Improved clock frequency compared to non-pipelined version
- Slight increase in register usage

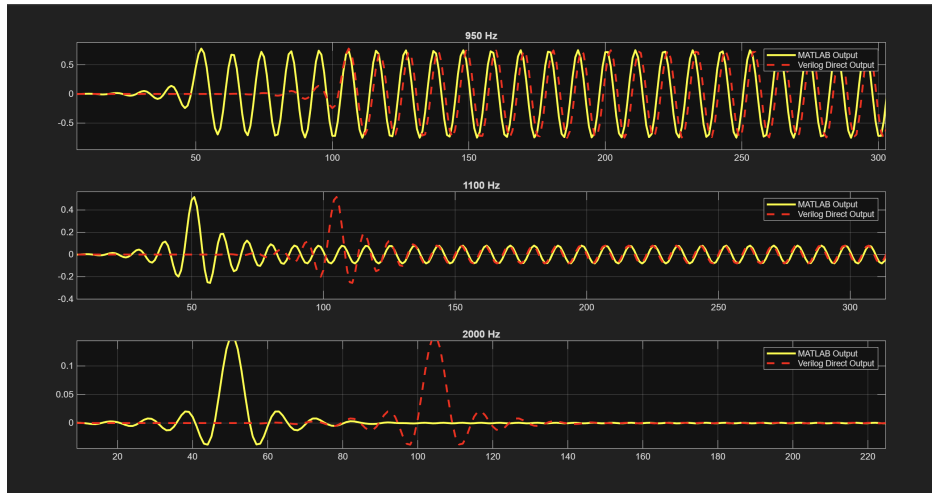


Figure 8: Optimized FIR Output (50 - stage Pipelining)

Resource Comparison (Optimized Form):

Flow Summary	
<<Filter>>	
Flow Status	In progress - Sat Apr 4 19:01:14 2026
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	exp5
Top-level Entity Name	fir_optimized
Family	Cyclone V
Device	5CGTFFD5C5F27I7
Timing Models	Final
Logic utilization (in ALMs)	N/A
Total registers	1616
Total pins	49
Total virtual pins	0
Total block memory bits	0
Total DSP Blocks	100
Total HSSI RX PCSs	0
Total HSSI PMA RX Deserializers	0
Total HSSI TX PCSs	0
Total HSSI PMA TX Serializers	0
Total PLLs	0
Total DLLs	0

Figure 9: Resource Utilization Comparison: Non pipelined

Flow Status	Successful - Wed Mar 25 22:00:10 2026
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	exp5
Top-level Entity Name	fir_optimized
Family	Cyclone V
Device	5CGTFFD5C5F27I7
Timing Models	Final
Logic utilization (in ALMs)	N/A
Total registers	45408
Total pins	66
Total virtual pins	0
Total block memory bits	0
Total DSP Blocks	50
Total HSSI RX PCSs	0
Total HSSI PMA RX Deserializers	0
Total HSSI TX PCSs	0
Total HSSI PMA TX Serializers	0
Total PLLs	0
Total DLLs	0

Figure 10: Resource Utilization Comparison: Pipelined

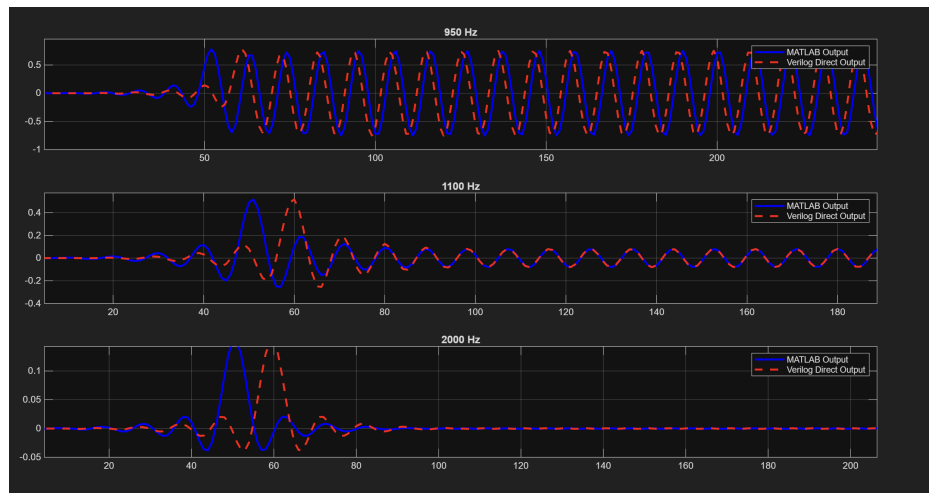


Figure 11: Optimized FIR Output (Tree Based Pipelining)

Resource Comparison (Optimized Form):

Flow Summary	
<<Filter>>	
Flow Status	In progress - Sat Apr 4 19:01:14 2026
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	exp5
Top-level Entity Name	fir_optimized
Family	Cyclone V
Device	5CGTFFD5C5F2717
Timing Models	Final
Logic utilization (in ALMs)	N/A
Total registers	1616
Total pins	49
Total virtual pins	0
Total block memory bits	0
Total DSP Blocks	100
Total HSSI RX PCSs	0
Total HSSI PMA RX Deserializers	0
Total HSSI TX PCSs	0
Total HSSI PMA TX Serializers	0
Total PLLs	0
Total DLLs	0

Figure 12: Resource Utilization Comparison: Non pipelined

Flow Summary	
<<Filter>>	
Flow Status	In progress - Mon Apr 13 03:52:10 2026
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	exp5
Top-level Entity Name	fir_optimized
Family	Cyclone V
Device	5CGTDFD5C5F2717
Timing Models	Final
Logic utilization (in ALMs)	N/A
Total registers	3514
Total pins	66
Total virtual pins	0
Total block memory bits	0
Total DSP Blocks	25
Total HSSI RX PCSs	0
Total HSSI PMA RX Deserializers	0
Total HSSI TX PCSs	0
Total HSSI PMA TX Serializers	0
Total PLLs	0
Total DLLs	0

Figure 13: Resource Utilization Comparison: Pipelined

Fmax Comparison (Optimized Form):

Slow 1100mV 100C Model Fmax Summary				
<<Filter>>				
	Fmax	Restricted Fmax	Clock Name	Note
1	51.53 MHz	51.53 MHz	clk	

This panel reports FMAX for every clock in the design, regardless of the user-specified clock periods. FMAX is only computed for paths where the source and destination registers or ports are driven by the same clock. Paths of different clocks, including generated clocks, are ignored. For paths between a clock and its inversion, FMAX is computed as if the rising and falling edges are scaled along with FMAX, such that the duty cycle (in terms of a percentage) is maintained. Altera

Figure 14: Fmax Comparison: Non pipelined

Slow 1100mV 100C Model Fmax Summary				
<<Filter>>				
	Fmax	Restricted Fmax	Clock Name	Note
1	102.11 MHz	102.11 MHz	clk	

Figure 15: Fmax Comparison: Pipelined

Multi Corner Timing Analysis Comparison (Optimized Form):

Multicorner Timing Analysis Summary						
<<Filter>>						
	Clock	Setup	Hold	Recovery	Removal	Minimum Pulse Width
1	▼ Worst-case Slack	25.156	0.117	N/A	N/A	19.178
1	clk	25.156	0.117	N/A	N/A	19.178
2	▼ Design-wide TNS	0.0	0.0	0.0	0.0	0.0
1	clk	0.000	0.000	N/A	N/A	0.000

Figure 16: Multi Corner Timing Analysis Comparison: Non pipelined

Multicorner Timing Analysis Summary						
<<Filter>>						
	Clock	Setup	Hold	Recovery	Removal	Minimum Pulse Width
1	▼ Worst-case Slack	5.457	0.033	N/A	N/A	6.678
1	clk	5.457	0.033	N/A	N/A	6.678
2	▼ Design-wide TNS	0.0	0.0	0.0	0.0	0.0
1	clk	0.000	0.000	N/A	N/A	0.000

Figure 17: Multi Corner Timing Analysis Comparison: Pipelined

3.3 Genvar-Based FIR

The Genvar-based FIR filter uses Verilog **generate** constructs to create scalable and parameterized hardware. This approach simplifies coding for large filter orders.

Key Features:

- Modular and scalable design

- Easy to modify number of taps
- Automatically generates repeated structures

Pipelined Genvar FIR:

Pipeline registers are introduced within the generated structure, typically inside loops, ensuring each stage is registered.

- Registers added inside generate blocks
- Uniform pipeline structure across taps
- Improved timing performance

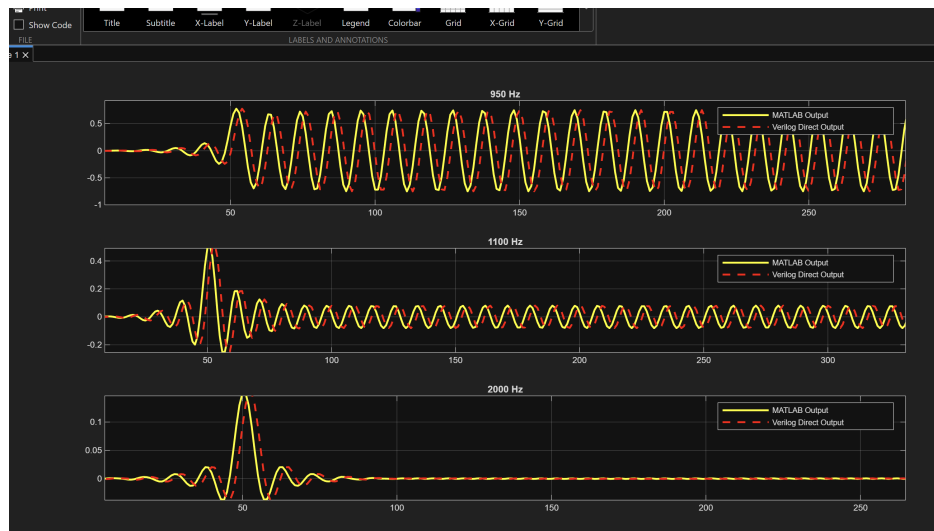


Figure 18: Genvar FIR Output (Pipelined)

Resource Comparison (Genvar Form):

Flow Summary	
<<Filter>>	
Flow Status	Successful - Thu Apr 9 12:26:59 2026
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	exp5
Top-level Entity Name	fir_genvar
Family	Cyclone V
Device	5CGTFD5C5F2717
Timing Models	Final
Logic utilization (in ALMs)	1,656 / 29,080 (6 %)
Total registers	1616
Total pins	66 / 364 (18 %)
Total virtual pins	0
Total block memory bits	0 / 4,567,040 (0 %)
Total DSP Blocks	99 / 150 (66 %)
Total HSSI RX PCSs	0 / 6 (0 %)
Total HSSI PMA RX Deserializers	0 / 6 (0 %)
Total HSSI TX PCSs	0 / 6 (0 %)
Total HSSI PMA TX Serializers	0 / 6 (0 %)
Total PLLs	0 / 12 (0 %)
Total DLLs	0 / 4 (0 %)

Flow Summary	
<<Filter>>	
Flow Status	Successful - Mon Apr 13 01:58:48 2026
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	exp5
Top-level Entity Name	fir_genvar
Family	Cyclone V
Device	5CGTFD5C5F2717
Timing Models	Final
Logic utilization (in ALMs)	3,964 / 29,080 (14 %)
Total registers	11099
Total pins	66 / 364 (18 %)
Total virtual pins	0
Total block memory bits	0 / 4,567,040 (0 %)
Total DSP Blocks	100 / 150 (67 %)
Total HSSI RX PCSs	0 / 6 (0 %)
Total HSSI PMA RX Deserializers	0 / 6 (0 %)
Total HSSI TX PCSs	0 / 6 (0 %)
Total HSSI PMA TX Serializers	0 / 6 (0 %)
Total PLLs	0 / 12 (0 %)
Total DLLs	0 / 4 (0 %)

Figure 19: Resource Utilization Comparison:Non Pipelined vs Pipelined

Fmax Comparison (Genvar Form):

Slow 1100mV -40C Model Fmax Summary				
<<Filter>>				
	Fmax	Restricted Fmax	Clock Name	Note
1	40.02 MHz	40.02 MHz	clk	

Slow 1100mV -40C Model Fmax Summary				
<<Filter>>				
	Fmax	Restricted Fmax	Clock Name	Note
1	143.8 MHz	143.8 MHz	clk	

Figure 20: Fmax Comparison:Non Pipelined vs Pipelined

Multi Corner Timing Analysis Comparison (Genvar Form):

Multicorner Timing Analysis Summary						
<<Filter>>						
	Clock	Setup	Hold	Recovery	Removal	Minimum Pulse Width
1	▼ Worst-case Slack	15.011	0.117	N/A	N/A	19.180
1	clk	15.011	0.117	N/A	N/A	19.180
2	▼ Design-wide TNS	0.0	0.0	0.0	0.0	0.0
1	clk	0.000	0.000	N/A	N/A	0.000

Multicorner Timing Analysis Summary						
<<Filter>>						
	Clock	Setup	Hold	Recovery	Removal	Minimum Pulse Width
1	▼ Worst-case Slack	8.046	0.116	N/A	N/A	6.678
1	clk	8.046	0.116	N/A	N/A	6.678
2	▼ Design-wide TNS	0.0	0.0	0.0	0.0	0.0
1	clk	0.000	0.000	N/A	N/A	0.000

Figure 21: Multi Corner Timing Analysis Comparison:Non Pipelined vs Pipelined

3.4 Pipeline Structure Explanation

Pipelining is a technique used to improve the operating frequency of digital circuits by inserting registers between combinational stages.

In FIR filters, pipelining is typically applied in:

- Multiplier outputs
- Adder tree stages

Effect of Pipelining:

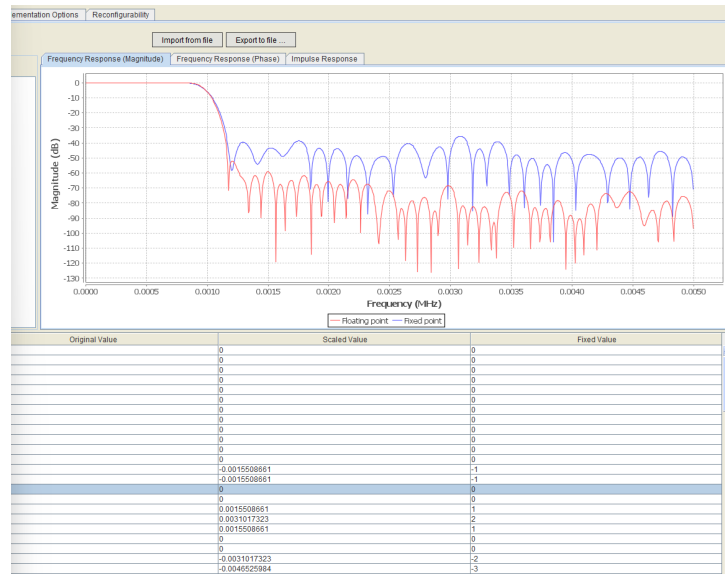
- Reduces critical path delay
- Increases maximum clock frequency (F_{max})
- Introduces latency (output delayed by pipeline depth)
- Increases register usage

3.5 FIR IP Core

The FIR IP core is generated using the Quartus FIR Compiler. It provides a highly optimized implementation using dedicated DSP blocks.

Features:

- Automatic pipelining
- Efficient DSP utilization
- High performance (maximum F_{max})



Resource Comparison with User Designs:

<<Filter>>	
Flow Status	Successful - Wed Mar 25 23:21:11 2026
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	exp5
Top-level Entity Name	firip
Family	Cyclone V
Device	5CGTFD5C5F27I7
Timing Models	Final
Logic utilization (in ALMs)	N/A
Total registers	207
Total pins	39
Total virtual pins	0
Total block memory bits	1,024
Total DSP Blocks	1
Total HSSI RX PCSs	0
Total HSSI PMA RX Deserializers	0
Total HSSI TX PCSs	0
Total HSSI PMA TX Serializers	0
Total PLLs	0
Total DLLs	0

4 Resource Utilization

4.1 Comparison Table

Parameter	Direct	Dir-Pipe	Opt	Opt-Pipe	Genvar	Gen-Pipe
Registers	1619	6384	45408	45408	1546	4767
DSP Blocks	99	100	50	50	99	100

Table 1: Resource Comparison of Designs

4.2 FIR IP

Parameter	FIR IP
Registers	207
DSP Blocks	1

5 Analysis

- Pipelining increases Fmax significantly for all architectures.
- Register count increases due to pipeline stages.
- Genvar design improves scalability and readability.
- Optimized design reduces redundant computations.
- FIR IP shows best timing performance due to hardware optimization.

6 Conclusion

- Pipelined architectures outperform non-pipelined ones in speed.
- Trade-off: increased registers vs improved timing.
- FIR IP provides best optimized solution.